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Apathy in Parkinson's disease: Why we should care

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The main objective of the study by Scott and colleagues, *Co-occurrence of apathy and impulse control disorder in Parkinson's disease* was to explore the possible co-existence of apathy and "impulse control disorders" (ICD) in a convenience sample of 887 Parkinson's disease (PD) patients (1). Definitions are important in this neurological study, as these behavioral entities are more often discussed in psychiatric journals than those devoted to neurology. The ICD's referenced in this study were generally compulsive rather than impulsive disorders in the purest sense. As such, the clinical issues of concern focused largely on repetitive and difficult to stop behaviors rather than "out of the blue" impulsive surges. The behaviors reported are typical of those included under this designation within neurological studies, but in the strictest sense they are more aligned with obsessive-compulsive behaviors than true impulsivity. In the same way, apathy has been variably defined, both as a symptom and as a syndrome. In this study it was defined by a score on a validated scale (2). The symptoms of apathy are diminished motivation, interest or emotional experience (2). The apathetic syndrome requires the addition of a variety of abnormalities associated with decreased goal directed behavior and the social consequences for these behavioral anomalies (2,3). The authors merge these strict distinctions for this study, but as the field expands further, they may need to be separated for clearer precision.

The article is anchored in a counterintuitive hypothesis that a lack of motivation or interest can co-exist with an irresistible urge to perform activities. For this reason alone, the observed association of apathy and ICD, even with the relatively loose definitions utilized, is of high

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clinical interest and has pertinence to clinicians and to those studying the neurochemical networks associated with such behaviors. In PD, the ICD's are among the non-motor "hyperdopaminergic" behaviors generally associated with the use of dopamine agonist drugs. Gambling is the most common ICD, but they include an array of peculiar (4) and generally very isolated compulsions and repetitive behaviors that unto themselves may not be abnormal, but become pathological because they cannot be interrupted. The relationship of apathy to dopamine is much less clear in PD. Apathy improves to a minor degree with dopamine agonists (3) and may be seen in non-motor "off" states (5) but is not generally part of the dopamine agonist withdrawal syndrome (6).

Scott et al (1) describe several correlations between apathy and other behaviors. The population was a highly selected population, most of whom had mini-mental state scores above 24 (non-demented), and were willing and able to fill out a computerized battery of tests by themselves. These results therefore do not generalize to the greater PD community, but that weakness does not detract from the results or their implications. These tests measured apathy, depression, anxiety, cognition, and ICD. Demographic data, including medications, were recorded as well. The authors found that 41.3% were apathetic, a number in keeping with other reports, and, to answer their main question, 17% of the total had significant symptoms of apathy while reporting at least one ICD. Hence, apathy and an ICD were not mutually exclusive. Use of dopamine agonists did not distinguish those with apathy from those with neither apathy or ICD. When those with depression were removed from the comparison, there were no significant changes in the correlations.

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Another counterintuitive finding was a positive correlation between apathy and anxiety. One might, a priori, assume that an apathetic person would not endorse symptoms of anxiety, since anxiety requires caring and concern whereas the hallmark of apathy is lack of emotion and motivation. All the rating instruments used had been validated in PD, and while our clinical experience has shown that patients often underrecognize apathy, (the patient says his life is fine but the spouse reports that, "He sits and does nothing all day long."), one study has shown that, once all definitions are clearly explained and the focus is on PD-related behaviors, caregivers and patients' self-ratings are surprisingly close. (7)

A single report, published while the Scott[1] article was under review (8) also examined the co-existence of apathy and compulsive behaviors, but in patients with primary obsessive-compulsive disorder without an underlying neurological disorder. They too found a positive correlation between these two behaviors, but this observation was thought to be mediated through depression, which was not the case in the current report on PD patients.

While the Scott et al results require confirmation, several take home messages are noteworthy. First, the notion of a linear spectrum of motivation, with apathy at one end and ICD's at the other is an oversimplification. Second, apathy is a problem that is common and serious in PD, not restricted to the demented and not clearly related to other hypodopaminergic behaviors. Finally, apathy may keep strange company, cohabiting with disorders of excess motivation, ICD's, and anxiety. Clinicians need to be aware of these counterintuitive associations to better understand a patient's and family's distress.

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As a final comment not specifically covered by the authors, the growing interest in non-motor behaviors is an opportunity for greater precision in definitions. If neurologists are entering these areas previously dominated by psychiatric colleagues, it may be a very good time for us to humbly study the well-considered definitions of behavioral traits. Just as neurologists educated psychiatrists that Extrapyramidal Syndrome was an obsolete ambiguity, we need to develop the clarity of expression to separate impulsions from compulsions, illusions from hallucinations, and a wide variety of terms if we wish to bring them into our neurological nosology with full precision.

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